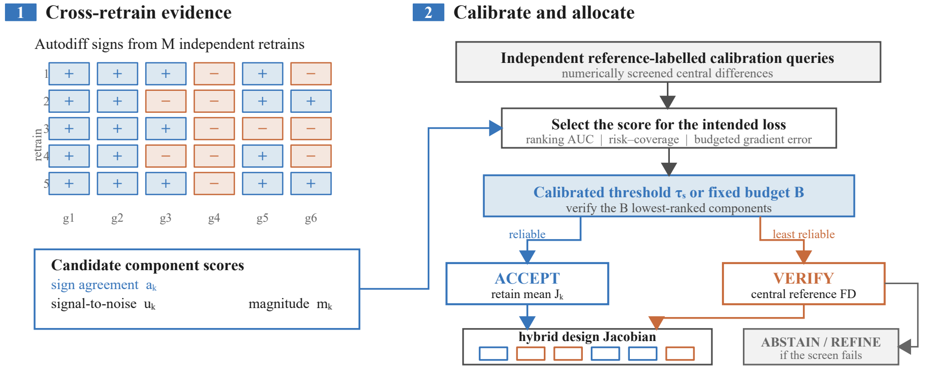
Reference-calibrated verification of metamodel design Jacobians

Rank components, verify only where calibrated evidence is insufficient, and report empirical risk at the actual simulation cost.



3 Report the operating point

External regimes at $\tau = 0.9$

benchmark	coverage	risk	evals
TMM near-saturated	95.9%	1.5%	0.90/22
TMM stressed	74.0%	13.4%	5.72/22
Heat/Poisson stressed	79.9%	4.9%	6.44/32

evals: selective / complete central FD

Score choice follows the deployment loss

sign-correctness AUC	**SNR / magnitude**
fixed-budget gradient error	**sign ordering**

3,580 / 3,580 external labels passed numerical screens

unresolved antenna derivatives were excluded

Structural boundary positive rescaling leaves sign agreement and SNR unchanged; shared bias can make every retrain agree and still be wrong

calibrated triage ≠ correctness certificate